

AI CHALLENGE FOR HEALTHY BRAIN AGING

PROPOSAL

PROPOSAL INFORMATION

Team Name: Neuronauts

Title of the Proposal: **DRISHTI** (Dementia Risk and Imaging-Subgroup Health Trajectory Index): A Harmonisation-Aware Multimodal Risk Model for MCI Conversion in Indian Adults

Team Members *	First Name	Last Name	Email ID	Institution Name
Lead Participant	Vijay	Soni	vs4vijay@gmail.com	Microsoft
AI Engineer	Ajay	Soni	as4ajaysoni@gmail.com	SYSTRA
ML Engineer	Vikram	Thula	vikramthula@microsoft.com	Microsoft

Declaration

- I and my team members hereby declare that we are **residing in India, affiliated to an Indian organisation.**
- I confirm that throughout the entire Challenge, I and my team members listed above, shall maintain complete confidentiality of all data, documents, materials, communications, and any other information shared with us in connection with the Challenge. We agree not to disclose, share, copy, or use this information for any purpose other than what is specifically allowed under the Challenge's terms and conditions.
- I understand that this obligation of confidentiality shall apply throughout the duration of the Challenge and, where applicable, continue thereafter in accordance with the governing policies and agreements. I also understand that any breach of this obligation may result in strict disciplinary or legal actions.
- I confirm that the information provided above is true and correct to the best of my knowledge and belief.

Date: 15/05/2026

Signature of the lead participant: Vijay Soni**

*Note: Only up to 5 members are allowed per team.

**Lead participant should declare on behalf of the entire team.

CHALLENGE AREA: Develop AI models and tools to advance and accelerate Alzheimer's and dementia research. Examples include prediction of development of mild cognitive impairment or dementia, models for disease progression, digital twins, data processing and harmonization tools etc.

DETAILED PROPOSAL (MAX 3 PAGES)

1. Abstract:

India is projected to host over 340 million adults aged 60+ by 2050, with a heavy cardio-metabolic disease burden that materially elevates dementia risk. Despite

this, Indians are systematically underrepresented in the global cohorts (ADNI, UK Biobank, OASIS, NACC, AIBL) that have shaped modern dementia AI. Risk models trained

on those cohorts transfer poorly to Indian populations, whose genetic background, education distribution, multilingual cognition, and vascular risk profile differ materially.

We propose DRISHTI, a multimodal risk model for Mild Cognitive Impairment (MCI) conversion, calibrated specifically on Indian longitudinal cohorts (TLSA, SANSCOG).

DRISHTI integrates cognitive assessments, MRI-derived features, OCT and OCT-Angio measures, blood biochemistry, and available omics through per-modality encoders and

a cross-modal fusion transformer, with a survival head producing time-to-MCI risk at 1, 3, and 5 year horizons. A cross-cohort harmonisation stack (neuroCombat plus

domain-adversarial training) lets us pretrain on global cohorts and fine-tune on the Indian data, closing the cohort-shift gap that has limited prior tools. Deep

ensembles and conformal prediction provide calibrated uncertainty intervals.

Expected outcomes: (a) an Indian-calibrated MCI risk model that meaningfully outperforms global-only baselines on TLSA and SANSCOG held-out splits; (b) a rigorous

subgroup fairness audit across sex, education, urban and rural strata, and age bands; (c) an open-source reference implementation plus a clinician-facing interactive

demo running inside the CBR secure environment; (d) a manuscript draft. The result is a tool, and a methodology, the Indian dementia research community can build on,

calibrated for the population it is meant to serve.

2. Project Overview:

The Lancet Commission on Dementia Prevention (2024) attributes up to 45 per cent of dementia cases to modifiable risk factors. India's modifiable burden, hypertension, diabetes, dyslipidemia, central obesity, hearing loss, and unequal education access, is exactly where intervention is most cost-effective. The clinical bottleneck is not the absence of risk factors. It is the absence of trustworthy, Indian-calibrated tools that tell a clinician which individuals are most likely to

convert to MCI within an actionable horizon. Existing global tools systematically miscalibrate on Indian patients, which both blunts prevention efforts and erodes clinician trust in AI-assisted risk stratification.

Objectives.

- *Train a multimodal MCI conversion model on TLISA and SANSCOG, harmonised against ADNI, OASIS, NACC, and AIBL.*
- *Quantify cohort-shift closure attributable to harmonisation, pretraining, and fine-tuning via ablation, treating this as a primary contribution rather than a footnote.*
- *Produce calibrated risk estimates at 1, 3, and 5 year horizons, with coverage of 80 per cent and 95 per cent prediction intervals within plus or minus 3 percentage points of nominal.*
- *Audit subgroup performance across sex, education, urban and rural, and age bands. No subgroup degraded more than 10 per cent from overall.*
- *Ship a clinician-facing demo running entirely inside the CBR secure environment, plus an open-source codebase to the extent data-use rules permit.*

Alignment with the Challenge. The Challenge call-out that 'Indians are underrepresented in such studies' is the core thesis of this proposal, not a footnote. The work addresses two of the four named in-scope examples (MCI prediction and data harmonisation tools) within a single coherent system, which we believe is the highest-impact and most credible use of the August 2026 to January 2027 execution window for a focused team.

3. AI Model Development Approach

Per-modality encoders (small MLPs for tabular cognitive, biochemistry, omics, OCT, and MRI-derived features) feed a cross-modal transformer fusion layer that produces a per-time-point latent representation. A survival head (discrete-time hazard or DeepSurv-style Cox-nn) produces time-to-MCI risk at 1, 3, and 5 year horizons, scored on concordance, AUROC, AUPRC, Brier score, and net benefit (decision-curve analysis). Subgroup-stratified versions of each metric are first-class outputs of every reported experiment, not optional add-ons.

Use of provided datasets (TLISA and SANSCOG). These are the primary training and validation cohorts. We consume the extracted and tabulated MRI volumetric and cortical-thickness measures, OCT and OCT-Angio retinal-layer and microvascular parameters, structured multi-domain cognitive assessments, full blood biochemistry, and available omics (ApoE genotype, methylation, proteomics). No raw DICOM access is required. Splits are participant-stratified to

prevent leakage across visits, with an external held-out site or wave subset reserved for cohort-shift evaluation.

External datasets. ADNI, OASIS, NACC, and AIBL are used solely to pretrain per-modality encoders, addressing the sample-size limits of Indian-only training. They are not used for headline metric reporting; final discrimination, calibration, and fairness claims are reported on TLSA and SANSCOG held-out splits.

Harmonisation stack. Feature level: neuroCombat for MRI features, analogous ComBat-style correction for biochemistry panels to remove scanner and site effects.

Representation level: a gradient-reversal head predicts cohort-of-origin from the fused representation, trained adversarially so the representation becomes cohort-invariant. Two-stage training: pretrain on the global pool, then fine-tune on TLSA and SANSCOG with a low learning rate and a regulariser that anchors the Indian-cohort head to the global head only where the data agrees.

Validation and evaluation. Deep ensembles (five seeds) for epistemic uncertainty; conformal prediction wraps the survival head for advertised coverage. Calibration plots, subgroup fairness audits, missing-modality stress tests, and decision-curve analyses are standard outputs of every reported experiment. Ablation studies isolate the contribution of harmonisation, pretraining, and fine-tuning, each as an independently reportable result.

4. Computing Resources Plan

CBR infrastructure (sealed surface). All training, evaluation, the validation report, and the jury-facing demo run inside the CBR secure remote environment per the Challenge's data-use rules. Requested resources: GPU access sufficient for transformer-fusion training over the order of 10,000 participants with order 100 tabular features and longitudinal observations. In practice this is one to two A100-class GPUs (or equivalent) for one to two weeks of cumulative training across ablation runs, plus standard CPU and RAM for tabular pipelines, harmonisation, and the demo app.

External resources (open surface). Pretraining on ADNI, OASIS, NACC, and AIBL is performed on cloud resources funded through (a) Microsoft Azure research credits accessible through the lead participant's affiliation and (b) NVIDIA academic GPU programmes or equivalent institutional credits. Only model weights and harmonisation transforms, never participant-level data, cross the boundary from the open surface into the CBR sealed surface.

The architecture is intentionally lean (small per-modality encoders, mid-sized fusion transformer, conformal wrappers) so that the binding constraint is data quality, not compute. Mixed-precision training, gradient checkpointing, and ensemble size capped at five keep total GPU-hours well within the budget above. The architecture has a documented scale-up path (deeper fusion, additional pretraining cohorts, additional heads for imaging-trajectory forecasting and counterfactual analysis) if more compute and team capacity become available, but the headline deliverables do not depend on it.

5. Team Expertise and Relevant Experience:

Neuronauts is a three-person team that has deliberately scoped DRISHTI to what we can credibly deliver in the six-month execution window.

Vijay Soni (Lead Participant), Microsoft. Software and AI engineer with hands-on experience building production-scale machine learning systems, multimodal data pipelines, and secure data-handling workflows. Will lead model architecture, multimodal fusion, training infrastructure, the secure-environment workflow, the clinician-facing demo, and overall project delivery.

Ajay Soni, SYSTRA. Software and AI Simulation Engineer with experience in large-scale data systems and engineering analytics. Will lead the data pipeline, the harmonisation stack, (neuroCombat plus adversarial training), the validation harness, and subgroup fairness instrumentation.

Vikram Thula, Microsoft. Software Engineer with ML Back. Will lead the end to end model training, fine tuning, evals and validation of overall accuracy.

We will seek the review / guide with SMEs post selection.

CV* OF ALL THE TEAM MEMBERS

- **Attached**

Vijay Soni

✉ vs4vijay@gmail.com | ☎ +91-9166276700 | 📍 Bangalore, India | 🔗 <https://github.com/vs4vijay>

Education

Govt. Engineering College

B. TECH IN COMPUTER SCIENCE & ENGINEERING

GPA: 7.0

Ajmer, India

July 2008 – May 2012

Board of Secondary Education

SENIOR SECONDARY IN PHYSICS, CHEMISTRY, AND MATHS

GPA: 7.4

Ajmer, India

Jul. 2007 – Jun. 2008

Board of Secondary Education

SECONDARY

GPA: 7.7

Ajmer, India

Jul. 2005 – Jun. 2006

Experience

Microsoft

SENIOR SOFTWARE ENGINEER

- Working on MS Teams Product, Building Services and Distributed Systems
- PR Copilot: Automated pull request troubleshooting using AI LLM RAG
- AutoMatch: Leveraged AI and LLM to reduce manual testing efforts, increasing efficiency and accuracy.
- PR Gate Framework: Developed a framework to automate pull request validation and integration
- Developed AI-driven solutions to assist teams in triaging and managing bugs effectively
- Filed a patent for bug categorization and prioritization

Bangalore, India

Oct. 2022 – Present

Morgan Stanley Capital International (MSCI Inc.)

SR. ASSOCIATE

- Part of core platform team responsible for creating common services across all product lines
- Integration with various products like BarraOne, ESG, Index, Beon etc
- Created Scheduling Microservice for scheduling tasks via REST APIs
- Worked on Shared Reporting Platform for Generating PDF Reports and Entitlement Service for Authentication and Authorization
- Code Reviews and helping team to follow standard practices

Pune, India

Oct. 2020 – Oct. 2022

Srijan Technologies Pvt. Ltd.

TECH. LEAD

Goa, India

Nov. 2015 – Oct. 2020

PolicyBazaar.com

SR. SOFTWARE ENGINEER

Gurgaon, India

Apr. 2015 – Nov. 2015

Metacube Software Pvt. Ltd.

SOFTWARE ENGINEER

Jaipur, India

Aug. 2012 – Apr. 2015

Skills

AI: LangChain, SemanticKernel, AutoGen, OpenAI, RAG, MCP, Embeddings, LLaMa
Backend: NodeJS, Java, Spring, GoLang, Express, Python (Flask, FastAPI), Ruby on Rails, Serverless / Lambda, Clojure (Basics)
Frontend: JavaScript (ES5, ES6), TypeScript, ReactJS, Angular, WebSockets (Socket.io), Ionic / PhoneGap, PWA, jQuery
Architectures: Microservices, Monoliths, Event Driven, Event Sourcing, CQRS, Orchestration and Choreography
Databases: MySQL, PostgreSQL, CosmosDB, Kusto, AWS RDS, MongoDB, Cassandra, Redis, Oracle
DevOps: Docker, Kubernetes, Azure (AppService, ADO, AKS, etc), AWS (EC2, S3, Lambda, SQS), Rancher, CI/CD Pipelines
IoT: Arduino, Raspberry Pi, ESP32 / ESP8266, NodeRED, NodeMCU
API: RESTful APIs, OpenAPI (Swagger), GraphQL, API Management, Apigee, Kong
Misc.: Bash, TCP/IP, OpenCV, RabbitMQ, Pandas, JWT, OAuth, Cryptography, Apps Script, WebRTC
Tools: npm, pip, ssh, maven/gradle, git, vim, uv, bun, ELK
Interests: Distributed Systems, Computer and Application Security, Startups, Blockchain and Cryptocurrencies

Projects

Microsoft Teams - Engineering Systems

MS TEAMS IS COLLABORATION PLATFORM THAT UNIFIES CHATS, ONLINE MEETINGS, DOCUMENT SHARING FOR ORGANIZATION.

TypeScript, NodeJS, Python, C#,
Azure DevOps, AI LLM, API
Microservices, Distributed Systems

OneMSCI Shared Services Engineering

MSCI CORE PLATFORM WHICH PROVIDES COMMON FEATURES LIKE AUTHENTICATION, ENTITLEMENT, REPORTING, SCHEDULING, NOTIFICATION ETC FOR VARIOUS MSCI PRODUCTS LIKE BARRAONE, ESG, INDEX, BLUEBOX ETC.

NodeJS, Java, Spring, TypeScript,
Microservices, Airflow, Redis,
ReactJS, Rancher K8S, Azure

Globe Telecom - SaaS API Platform

GLOBE TELECOM IS THE MAJOR PROVIDER OF TELECOMMUNICATIONS SERVICES IN THE PHILIPPINES. GLOBE PROVIDES VARIOUS APIS TO DEVELOPERS TO INTEGRATE THEIR APPS IN THE GLOBE ECOSYSTEM. WE HAVE BUILT THE PLATFORM WITH MICROSERVICE ARCHITECTURE AND API MONETIZATION ON AWS CLOUD.

Java, Spring Boot, NodeJS, GoLang,
ReactJS, Microservices, PostgreSQL,
Express, Apigee, Docker, AWS

<http://www.globelabs.com.ph/>

KubeDrop

KUBEDROP IS A CLOUD NATIVE, HYBRID CLOUD PLATFORM SOLUTION FOR WEB APPLICATIONS. OPTIMIZED FOR DECOUPLED, HEADLESS & MICROSERVICES ARCHITECTURE AROUND CMS. SCALABLE TO MILLIONS OF PAGE REQUESTS ACROSS HUNDREDS OF SITES.

GoLang, NodeJS, ReactJS, Kong API
Gateway, Docker, Kubernetes, Kafka

Internet of Clean - Diversey

THE INTERNET OF CLEAN (IoC) INITIATIVE OF DIVERSEY PROVIDES A PLATFORM WHICH REMOTELY MONITORS THE EQUIPMENTS, MACHINES, AND OPERATIONS THROUGH SENSOR GENERATED DATA. THE PLATFORM FEATURES A DIGITAL PORTAL AND INTUITIVE DASHBOARDS WHERE USERS CAN REVIEW DATA AND TRENDS.

NodeJS, AngularJS, Python, MySQL,
TCP Server, IoT Devices, Cassandra

<https://diversey.com/>

Vega

VEGA IS AN INTERNAL TOOL BUILT FOR PROJECT SCAFFOLDING TO SPEED UP THE DEVELOPMENT PROCESS WITHIN SRIJAN. VEGA IS OPEN SOURCED UNDER SRIJANLABS ([HTTPS://GITHUB.COM/SRIJANONE/VEGA](https://github.com/srijanone/vega)).

GoLang

<https://github.com/srijanone/vega>

HealthCare at Home - HCAH

HEALTHCARE AT HOME (HCAH) IS A HOME HEALTHCARE SERVICE PROVIDER. IT PROVIDES ICU AT HOME, CANCER CARE AT HOME, NURSING CARE AT HOME, WITH MEDICINE DELIVERY FACILITY.

Ruby on Rails, NodeJS, Sinatra,
MySQL, MongoDB, AWS EC2

<https://hcah.in/>

PolicyBazaar.com

AN INSURANCE AGGREGATOR PLATFORM FOR CARS AND TWO WHEELERS FOR SEARCHING AND COMPARING BEST QUOTES FROM ALL INSURERS IN INDIA ACCORDING TO CUSTOMER NEEDS. CUSTOMERS CAN COMPARE AND BUY POLICIES ONLINE AND MAKE PAYMENTS THROUGH VARIOUS PAYMENT GATEWAYS IN INDIA.

Java, Spring, AngularJS, NodeJS,
Web Services, MongoDB

<https://policybazaar.com/>

Metacampus

METACAMPUS IS ADVANCED LEARNING PLATFORM THAT ALLOWS MULTIPLE STREAMS OF COMMUNICATION BETWEEN PARENTS, TEACHERS AND STUDENTS. ANDROID AND IOS APPS ARE ALSO PROVIDED FOR USER.

Java, AngularJS, Spring, Ionic,
Google BigTable, Google App Engine

<https://metacampus.in>

Awards

2025	Patent , System for Context-Aware Bug Classification and Prioritization for Enterprise Applications	USPTO
2022	Microsoft Certified , Azure AI Fundamentals, Azure Security Associate, Azure Developer Associate	Microsoft
2009-13	Linux For You Magazine , Got tips published in Linux For You Magazine multiple times	
Jun. 2010	Winner , Winner of Linux Whiz at Neuron'10 held by MNIT (Jaipur)	MNIT, Jaipur
Jul. 2011	Runner Up , Runner up at Nation Level Ethical Hacking Contest (InCTF'11)	InCTF
Sept. 2014	Runner Up , Secured 2nd Position at Adobe Security Challenge	Adobe
Mar. 2015	Runner Up , Runner up at Google Chrome Extension challenge for creating CalEvent	DevPost

Ajay Soni

Senior Engineer – Simulation

Delhi/NCR

+91 8764160444

as4ajaysoni@gmail.com

<https://www.linkedin.com/in/as4ajaysoni/>

TECHNICAL SKILLS

AI: RAG, Embeddings

Simulation Tools: OpenFOAM, Subway Environment Simulation (SES), Camatt 2.20, Subway Ventilation Simulation (SVS), FDS/SMV, Pyrosim, ANSYS Fluent, EnergyPlus, eQuest, Pathfinder, Paraview, Salome

Design softwares: AutoCAD, CATIA, SolidWorks

Standards familiar: NFPA, Eurovent, CIBSE, ASHRAE, ISHRAE, ARI, ISO, AHRI

Programming Languages: Python, Excel VBA (Visual Basic for Applications), MATLAB, BASH, C/C++

IoT: Arduino, Raspberry Pi, ESP32 / ESP8266

Tools: npm, pip, ssh, git, vim, uv, bun

Operating System: Linux, Windows

PROFESSIONAL WORK EXPERIENCE

SYSTRA MVA Consulting (India) Private Limited, Faridabad -Mechanical Engineer

JULY 2019 - PRESENT (6 Years)

- Dubai Metro Blue Line (DMBL) – Involve the development of a 1-D Subway Ventilation Simulation (SVS) model for normal, congested, and emergency scenarios. The model was used to design the tunnel ventilation system, including the specification of tunnel ventilation fans, jet fans, and cooling requirements. 3D CFD Model is developed to find stratification of temperature inside tunnel. (1/2025 - present)
- Patna Metro Corridor-1 & Corridor-2 – Responsible for performing CFD Simulation of temperature evolution around a stopped train in the tunnel during congestion using Fire Dynamics Simulator (FDS). Developed 1-D SVS (Subway Ventilation Simulation) model for Normal, Congested and Emergency scenarios. Various iterations are performed to determine heat load for 7 Underground stations of Corridor-1 and 6 Underground Station of Corridor-2 determined due to train movement in normal operation. Client: DMRC (1/2021 - present)

Development of Performance evaluation method for composite Air Handling Unit (AHU) in Indian Context

M.Tech.

Project

JULY 2018 - JUNE 2019

CFD Analysis of Internal Combustion Engine

B.Tech. Project

JULY 2014 - JUNE 2015

EDUCATION

NIT Jaipur (MNIT)

- M.Tech. in Thermal Engineering

JUL 2017 - JUN 2019

Govt. Engineering College Ajmer

- B.Tech. in Mechanical Engineering

JUL 2011 - JUN 2015

AWARDS & PUBLICATIONS

- **Standard Published:** "Standard for Air Handling Units: General Requirements, Performance testing and Rating". ISHRAE – RAMA Standard – 20002: 2020
- **Semifinal Winner of aQuest 2019:** aQuest is a competition organised by ISHRAE and Powered by HITACHI every year, which is concluded with Grand Finale at ACREX INDIA, South Asia's largest Exhibition on Refrigeration and Cold Chain, Air-Conditioning, Ventilation and Intelligent Buildings Services
- **Secured 6th rank in CMTR:** Secured 6th rank in CMTR (College Level Mathematics Talent Hunt) in all over of Rajasthan
- **Membership:** Member of ISHRAE-RAMA Standards Development committee, ASHRAE India Chapter, ISHRAE Jaipur Chapter

VIKRAM THULA

Phone: +91 9393039546

E-mail: Vikram.thula@outlook.com

Career Objective

Experienced R&D and Innovation Lab Engineer with over 12 years of hands-on expertise in prototyping, lab operations, and technical mentoring. Currently managing The Microsoft Garage Lab (via DXC), delivering impactful support across Azure IoT, robotics, VR, and embedded platforms. Known for building and scaling innovative environments, conducting global hackathons, and driving high-impact Proof of concepts across industries. Certified in Azure Fundamentals and DJI Enterprise systems.

Professional Experience & Responsibilities

Data center Lab Manager – Microsoft

Hyderabad (Sep 2025 – present)

- specializing in hardware lab operations, Servers bring-up, validation, instrumentation, and test automation within the Data center Labs.
- Hands-on experience with lab infrastructure, debugging the hardware-software interface, telemetry/log analysis, and building reliable test pipelines for large-scale device deployments.
- Working knowledge of Python for data processing and automation, with exposure to ML workflows — data preprocessing, feature engineering, model training/evaluation — and an interest in signal processing for biomedical applications (EEG, MRI, speech, gait).

Lab Manager – Microsoft, The Garage (via DXC)

Bengaluru (May 2021 – Sep 2025)

- Drive lab operations include equipment maintenance, safety, security, audits, procurement, and inventory for The Garage Makerspace, Advanced Lab, and VR Lab.
- Identify infrastructure, hardware, consumables, and technical requirements to support new project implementations across multiple business units.
- Lead system integration for emerging hardware/software, Azure IoT PoCs, and lab-based innovation prototypes; ensure compliance and safety protocols.
- Own end-to-end execution of Global Hackathon events (pre/during/post), including training sessions, logistics, and stakeholder collaboration.
- Manage lab service requests, vendor interfaces, risk assessments, and escalations to ensure seamless operations and project continuity.
- Communicate project schedules, dependencies, and risks to stakeholders while ensuring high-quality execution and continuous improvement of lab experiences.

Senior R&D Engineer – Experience Center - Cyient Ltd.

Hyderabad (May 2016 – May 2021)

- System configurations and graphical interfaces for Linux in Open-source boards like Beagle board Black, Jetson TK1.
- ROS on the Jetson TK1 and Linux custom boards.
- Manage Experience Center operations from maintaining all lab equipment to logistics of lab consumables and facilitate all technical matters with BUs on IoT, UAV, Robotics, LEGO, AR / VR, and Ansys Labs.
- Assists and guide associates from different business units to formulate their ideas for prototype models.
- Interact and demonstrate proof of concepts to international clients and external visitors of Experience Center.
- End to end Drone practitioner - building, integrating, calibrating sensors, flying, repairing of Drone, Data acquisition, processing, reporting and presenting into 3D models and Maps.
- Integrating open hardware platforms like Arduino and Jetson, with various sensors based on the application demand.
- Coached Internal and External Hackathons
- Technical Mentor in **HackADrone** to take part in DJI SDK and open hardware Integrations to Enterprise Drone Platforms.
- Worked as a Hardware Lab Configuration Engineer on the Hitachi STS Metro project, using Azure DevOps and CI/CD practices for system integration and automation.
- Developed various PoC's in Drones, IoT, Robotics, and LEGO.

B. Tech.	Electronics & Instrumentation Engineering	Keshav Memorial Institute of Technology, Hyderabad	2007-2011
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Achievements

- **Microsoft Certified:** Azure Fundamentals have demonstrated foundational level knowledge of cloud services and how those services are provided with Microsoft Azure [verification link](#)
- **Directorate General of Civil Aviation, Government of India (DGCA)** certified Remote Drone/UAV Pilot [verification link](#)
- **Pix4D** Certified Mapper - Drone data Mapping software [verification link](#)
- **DJI** authorized service engineer for DJI Enterprise platforms
- Published an IEEE paper “Wireless Data Acquisition System for Vital sign Monitoring of Neonates” [verification link](#)
- Successfully completed the certificates below.
 - **6.002x: Circuits and Electronics**, the course offered by **THE MASSACHUSETTS INSTITUTE OF TECHNOLOGY** [verification link](#)
 - **EE40LX: Electronic Interfaces: Bridging the Physical and Digital Worlds**, the course offered by **Berkeley** [verification link](#)
 - **LFS101x: Introduction to Linux**, the course offered by **Linux Foundation** [verification link](#)
 - **CloudIntro.x, Introduction to Cloud Computing**, an online learning initiative course offered by **IEEE X** [verification link](#)
 - **Stroke201x: Science of Stroke Recovery** [verification link](#)
 - **ANA101x: Human Anatomy for Stroke** [verification link](#)